# Signals and Systems Lecture-14: Laplace Transform 1

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# Introduction (1)

Consider a continuous-time signal $x(t)$. The Laplace transform of $x(t)$ is defined as:

$$X(s) = \mathcal{L}[x(t)] = \int_{-\infty}^{+\infty} x(t)e^{-st}\mathrm{d}t$$

- $s$ is a complex number:

$$\begin{aligned} s &= \mathrm{Re}(s) + j\mathrm{Im}(s) \\ &\triangleq \sigma + j\omega \end{aligned}$$

- The Fourier transform expresses $x(t)$ as a linear combination of complex sinusoids of the form $e^{j2\pi ft} = e^{j\omega t}$.
- The Laplace transform expresses $x(t)$ as a linear combination of complex exponentials of the form $e^{st}$.

# Introduction (2)

**Proposition:** $X(s)$ is the Fourier transform of $x(t)e^{-\sigma t}$.
*Proof:*

- The proof follows from the definition of $X(s)$:

$$\begin{aligned} X(s) = \mathcal{L}[x(t)] &= \int_{-\infty}^{+\infty} x(t)e^{-st}\mathrm{d}t \\ &= \int_{-\infty}^{+\infty} x(t)e^{-(\sigma+j\omega)t}\mathrm{d}t \\ &= \int_{-\infty}^{+\infty} \left[x(t)e^{-\sigma t}\right] e^{-j\omega t}\mathrm{d}t \\ &= \mathcal{F}\left[x(t)e^{-\sigma t}\right] \end{aligned}$$

where $\omega = 2\pi f$.

# Introduction (3)

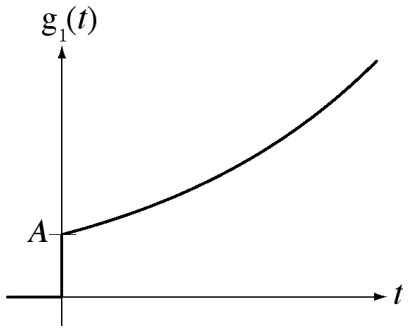
Consider the case where the signal $x(t)$ grows with time (example: $x(t) = e^t$). For this kind of signals:

- It is impossible to calculate the Fourier transform of $x(t)$ since the integral will not converge.
- Since $X(s) = \mathcal{F}\left[x(t)e^{-\sigma t}\right]$:
  - By an appropriate choice of $\sigma = \mathrm{Re}(s)$, $x(t)e^{-\sigma t}$ can be made a decreasing function of time.
  - In this case, it is possible to calculate the Fourier transform of $x(t)e^{-\sigma t}$ (and consequently the Laplace transform of $x(t)$).
  - $e^{-\sigma t}$ is called the convergence factor.
    - When chosen properly, it makes the integral converge for some signals when it would not otherwise converge.
- Unlike the Fourier transform, the Laplace transform is defined by:
  1. The algebraic expression of the transform ($X(s)$).
  2. The Region Of Convergence (ROC).

# Region Of Convergence (1)

Calculate the Laplace transform of:

$$g_1(t) = Ae^{\alpha t}u(t) \;\; ; \;\; \alpha > 0$$




$$G_1(s) = \int_{-\infty}^{+\infty} g_1(t)e^{-st}\mathrm{d}t = A\int_{0}^{+\infty} e^{(\alpha-s)t}\mathrm{d}t$$

$$= \frac{A}{s-\alpha}\left[1 - \lim_{t\to+\infty} e^{(\alpha-s)t}\right] = \frac{A}{s-\alpha}$$

where the last equation is verified if $e^{(\alpha-s)t} \to 0$ as $t \to +\infty$:

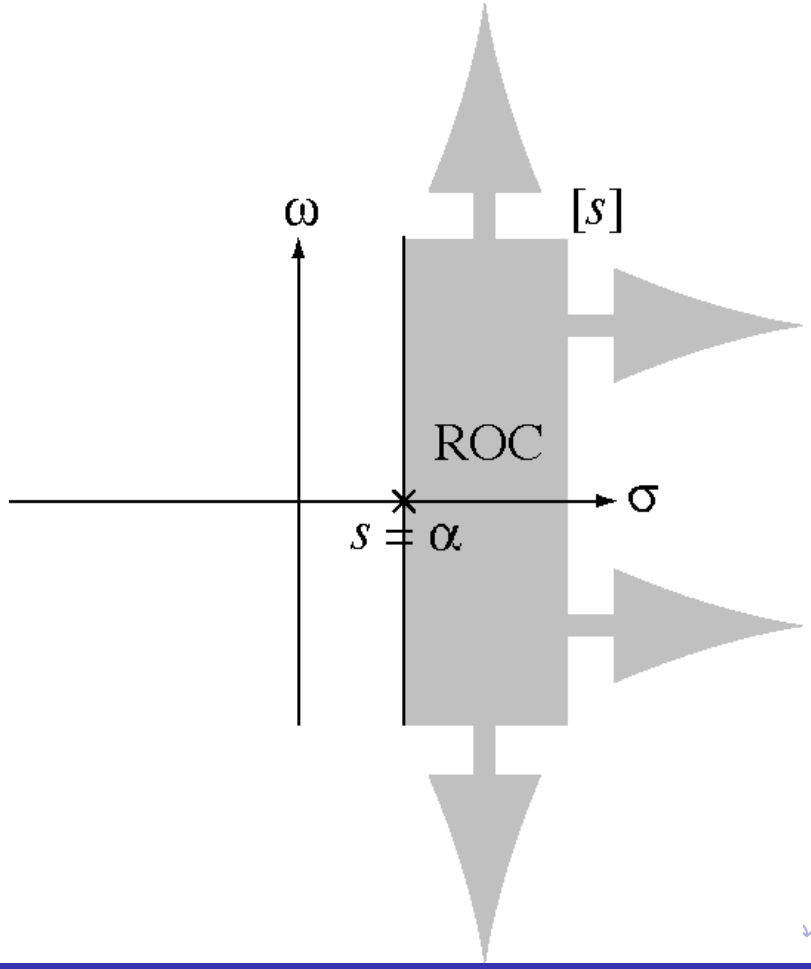
$$\left|e^{(\alpha-s)t}\right| = \left|e^{(\alpha-\sigma)t}\right| \underbrace{\left|e^{-j\omega t}\right|}_{=1 \;\; \forall\, \omega,t}$$

# Region Of Convergence (2)

- Consequently, the integral will converge only when $\alpha - \sigma < 0$ since in this case:

$$\left|e^{(\alpha-s)t}\right| = \left|e^{(\alpha-\sigma)t}\right| \to 0 \text{ as } t \to +\infty$$

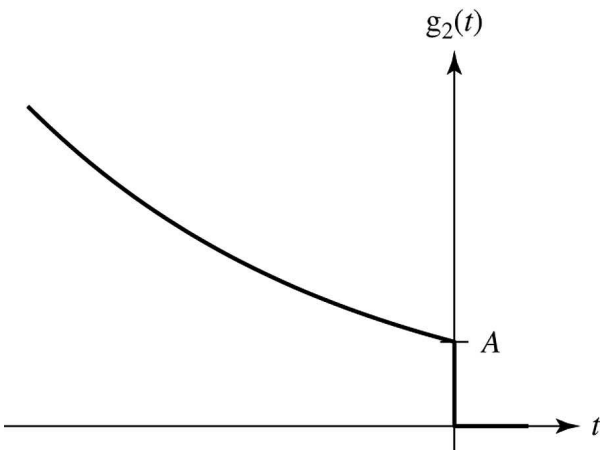
- Therefore, the Laplace transform of $g_1(t)$ is defined by:
  - $G_1(s) = \frac{A}{s-\alpha}$.
  - The ROC is given by $\sigma = \mathrm{Re}(s) > \alpha$.



# Region Of Convergence (3)

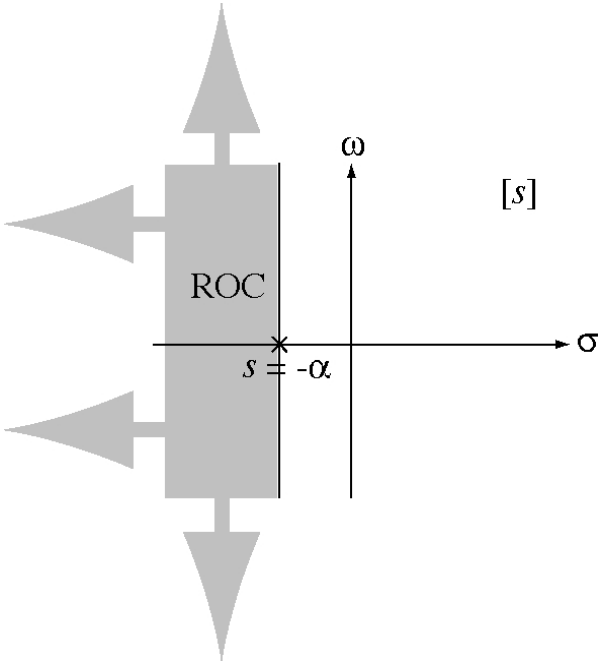
Consider the function given by:

$$g_2(t) = Ae^{-\alpha t}u(-t)\ ;\ \alpha > 0$$




By a similar reasoning, we can show that:

- $\mathcal{L}\left[g_2(t)\right]$ is given by:
  - $G_2(s) = -\frac{A}{s+\alpha}$.
  - The ROC is given by $\sigma = Re(s) < -\alpha$.



# Region Of Convergence (4)

The following two Laplace transforms illustrate the importance of the region of convergence.

$$e^{-\alpha t}u(t) \leftrightarrow \frac{1}{s+\alpha} \;\; ; \;\; ROC = \{s \mid \sigma = \mathrm{Re}(s) > -\alpha\}$$

$$-e^{-\alpha t}u(-t) \leftrightarrow \frac{1}{s+\alpha} \;\; ; \;\; ROC = \{s \mid \sigma = \mathrm{Re}(s) < -\alpha\}$$

The two time-domain functions are different but:

- The algebraic expressions for their Laplace transforms are the same.
- Only the ROCs are different.

# Relation between the FT and the LT

The Laplace transform is the Fourier transform with $j\omega = j2\pi f$ replaced by $s$ **only** when the ROC includes the $j\omega$ axis.

- For example:

$$e^{-\alpha t}u(t) \leftrightarrow \frac{1}{s+\alpha} \;\; ; \;\; ROC = \{s \mid \sigma = \mathrm{Re}(s) > -\alpha\}$$

  - When $\alpha > 0$, then the ROC includes the $j\omega$ axis.
  - In this case, the Fourier transform is given by:

$$\mathcal{F}\left[e^{-\alpha t}u(t)\right] = \frac{1}{\alpha + j2\pi f}$$

  - When $\alpha < 0$, the ROC does not include the $j\omega$ axis and $e^{-\alpha t}u(t)$ does not have a Fourier transform.

# Unilateral Laplace Transform (1)

- In what follows, we will be interested in causal signals. This is the most common application for Laplace transforms.
- The bilateral Laplace transform of a causal signal $x(t)u(t)$ is given by:

$$X(s) = \int_{-\infty}^{+\infty} x(t)u(t)e^{-st}\mathrm{d}t$$
$$= \int_{0^-}^{+\infty} x(t)e^{-st}\mathrm{d}t$$

  - This is the unilateral Laplace transform.
  - The lower limit $0^-$ indicates that we include impulses at the origin.
- In what follows, we will be only studying the unilateral Laplace transform which will be referred to as the Laplace transform.

# Unilateral Laplace Transform (2)

- A function and its Laplace transform are uniquely related only if the function is causal.
- Any function which grows no faster than an exponential in positive time has a unilateral Laplace transform.

# Example-1

Calculate the Laplace transform of $u(t)$:

$$U(s) = \int_{0^-}^{+\infty} u(t)e^{-st}\mathrm{d}t = \int_{0^-}^{+\infty} e^{-st}\mathrm{d}t$$
$$= -\frac{1}{s}\left[e^{-st}\right]_{0^-}^{+\infty}$$

- $U(s) = \frac{1}{s}$ provided that $e^{-st} \to 0$ as $t \to +\infty$:

$$\left|e^{-st}\right| = \left|e^{-\sigma t}\right| \underbrace{\left|e^{-j\omega t}\right|}_{=1 \ \ \forall\ \omega,t}$$

  - Consequently, the ROC is defined by:

$$ROC = \{s \mid \sigma = \mathrm{Re}(s) > 0\}$$

- Note that the $j\omega$ axis is not included in the ROC and the Fourier transform of $u(t)$ is <u>not</u> equal to $\frac{1}{j2\pi f}$.
  - In fact, we have previously seen that the Fourier transform of $u(t)$ is $\frac{1}{j2\pi f} + \frac{1}{2}\delta(f)$.

# Example-2 (1)

Calculate the Laplace transform of the function:

$$x(t) = e^{-(\alpha+j\beta)t}u(t)$$

where $\alpha$ and $\beta$ are real numbers.

- The Laplace transform is given by:

$$\begin{aligned} X(s) &= \int_{0^-}^{+\infty} e^{-(\alpha+j\beta)t}u(t)e^{-st}\mathrm{d}t \\ &= \int_{0^-}^{+\infty} e^{-(s+\alpha+j\beta)t}\mathrm{d}t \\ &= -\frac{1}{s+\alpha+j\beta}\left[e^{-(s+\alpha+j\beta)t}\right]_{0^-}^{+\infty} \\ &= \frac{1}{s+\alpha+j\beta}\left[1-\lim_{t\to+\infty} e^{-(s+\alpha+j\beta)t}\right] = \frac{1}{s+\alpha+j\beta} \end{aligned}$$

where the last equation holds under the condition that: $e^{-(s+\alpha+j\beta)t} \to 0$ as $t \to +\infty$.

# Example-2 (2)

- On the other hand:

$$\left|e^{-(s+\alpha+j\beta)t}\right| = \left|e^{-(\sigma+\alpha)t}\right| \underbrace{\left|e^{-j(\omega+\beta)t}\right|}_{=1}$$

- Consequently, the ROC corresponds to the values of $s$ verifying the relations: $\sigma + \alpha > 0$.
- Finally:

$$e^{-(\alpha+j\beta)t}u(t) \leftrightarrow \frac{1}{s+(\alpha+j\beta)} \quad ; \quad ROC = \{s \mid \sigma = \mathrm{Re}(s) > -\alpha\}$$

# Example-3

Calculate the Laplace transform of the damped sinusoid $e^{-\alpha t}\cos(\omega_0 t)u(t)$ $(\alpha \geq 0)$:

$$e^{-\alpha t}\cos(\omega_0 t)u(t) = e^{-\alpha t}\left[\frac{1}{2}e^{j\omega_0 t} + \frac{1}{2}e^{-j\omega_0 t}\right]u(t)$$

$$\begin{aligned}
\Rightarrow \mathcal{L}\left[e^{-\alpha t}\cos(\omega_0 t)u(t)\right] &= \mathcal{L}\left[\frac{1}{2}e^{-(\alpha - j\omega_0)t}u(t)\right] + \mathcal{L}\left[\frac{1}{2}e^{-(\alpha + j\omega_0)t}u(t)\right] \\
&= \frac{1}{2}\left[\frac{1}{s + (\alpha - j\omega_0)} + \frac{1}{s + (\alpha + j\omega_0)}\right] \\
&= \frac{s + \alpha}{(s + \alpha)^2 + \omega_0^2}
\end{aligned}$$

- Based on example 2, the ROCs of $e^{-(\alpha - j\omega_0)t}u(t)$ and $e^{-(\alpha + j\omega_0)t}u(t)$ correspond to $\sigma > -\alpha$. Consequently:

$$ROC = \{s \mid \sigma = \mathrm{Re}(s) > -\alpha\}$$

# Example-4 (1)

Calculate the Laplace transform of the damped sinusoid $e^{-\alpha t}\sin(\omega_0 t)u(t)$ $(\alpha \geq 0)$:

$$e^{-\alpha t}\sin(\omega_0 t)u(t) = e^{-\alpha t}\left[\frac{1}{2j}e^{j\omega_0 t} - \frac{1}{2j}e^{-j\omega_0 t}\right]u(t)$$

$$\Rightarrow \mathcal{L}\left[e^{-\alpha t}\sin(\omega_0 t)u(t)\right] = \mathcal{L}\left[\frac{1}{2j}e^{-(\alpha - j\omega_0)t}u(t)\right] - \mathcal{L}\left[\frac{1}{2j}e^{-(\alpha + j\omega_0)t}u(t)\right]$$

$$= \frac{1}{2j}\left[\frac{1}{s+(\alpha - j\omega_0)} - \frac{1}{s+(\alpha + j\omega_0)}\right]$$

$$= \frac{\omega_0}{(s+\alpha)^2 + \omega_0^2}$$

- Based on example 2, the ROCs of $e^{-(\alpha - j\omega_0)t}u(t)$ and $e^{-(\alpha + j\omega_0)t}u(t)$ correspond to $\sigma > -\alpha$. Consequently:

$$ROC = \{s \mid \sigma = \mathrm{Re}(s) > -\alpha\}$$

# Example-4 (2)

- Setting $\alpha = 0$ in example-3 and example-4 results in:

$$\cos(\omega_0 t)u(t) \leftrightarrow \frac{s}{s^2 + \omega_0^2}$$
$$\sin(\omega_0 t)u(t) \leftrightarrow \frac{\omega_0}{s^2 + \omega_0^2}$$

- The ROC is given by the values of $s$ verifying $\mathrm{Re}(s) > 0$.
- Note that for the sine and cosine functions, the Laplace transform does not have the same form as the Fourier transform since the $j\omega$ axis is not in the ROC.

# Example-5

Calculate the Laplace transform of $\delta(t)$:

$$\mathcal{L}\left[\delta(t)\right] = \int_{0^-}^{+\infty} \delta(t) e^{-st} \mathrm{d}t$$
$$= \int_{0^-}^{+\infty} \delta(t) e^{-s0} \mathrm{d}t = \int_{0^-}^{+\infty} \delta(t) \mathrm{d}t = 1 \ \ \forall\, s$$

Since the $j\omega$ axis is included in the ROC, the Fourier transform and the Laplace transform are similar:

$$\delta(t) \rightleftharpoons \mathcal{F}\left[\delta(t)\right] = 1$$
$$\delta(t) \leftrightarrow \mathcal{L}\left[\delta(t)\right] = 1$$

# Summary

To summarize:

$$
\begin{aligned}
e^{-\alpha t}u(t) &\leftrightarrow \frac{1}{s+\alpha} && \mathrm{Re}(s) > -\alpha \\
u(t) &\leftrightarrow \frac{1}{s} && \mathrm{Re}(s) > 0 \\
\cos(\omega_0 t)u(t) &\leftrightarrow \frac{s}{s^2+\omega_0^2} && \mathrm{Re}(s) > 0 \\
\sin(\omega_0 t)u(t) &\leftrightarrow \frac{\omega_0}{s^2+\omega_0^2} && \mathrm{Re}(s) > 0 \\
\delta(t) &\leftrightarrow 1 && \forall \ s
\end{aligned}
$$

# Some Properties of the Laplace Transform (1)

- Linearity:

$$\alpha x(t) + \beta y(t) \leftrightarrow \alpha X(s) + \beta Y(s)$$

- Time Shifting:

$$x(t - t_0) \leftrightarrow X(s)e^{-st_0} \quad ; \quad t_0 \geq 0$$

Note that, we must have $t_0 \geq 0$ so that $x(t - t_0)$ will remain causal whenever $x(t)$ is causal.

- Complex-Frequency Shifting:

$$e^{s_0 t}x(t) \leftrightarrow X(s - s_0)$$

- Multiplication-Convolution Duality:

$$x(t) * y(t) \leftrightarrow X(s)Y(s)$$
$$x(t)y(t) \leftrightarrow X(s) * Y(s)$$

# Some Properties of the Laplace Transform (2)

- Time Scaling:

$$x(at) \leftrightarrow \frac{1}{a} X\left(\frac{s}{a}\right) \quad ; \quad a > 0$$

  Note that, we must have $a > 0$ so that $x(at)$ will remain causal whenever $x(t)$ is causal.

- Complex-Frequency Scaling:

$$\frac{1}{a} x\left(\frac{t}{a}\right) \leftrightarrow X(as) \quad ; \quad a > 0$$

  Note that, we must have $a > 0$ so that $x(\frac{t}{a})$ will remain causal whenever $x(t)$ is causal.

Note that the above properties are very similar to the properties of the Fourier transform.

# Example-6

Calculate the Laplace transforms of $e^{-\alpha t}\cos(\omega_0 t)u(t)$ and $e^{-\alpha t}\sin(\omega_0 t)u(t)$ ($\alpha$ is real).

- Applying the complex-frequency shifting property of the Laplace transform:

$$\cos(\omega_0 t)u(t) \leftrightarrow \frac{s}{s^2+\omega_0^2}$$

$$\Rightarrow \quad e^{-\alpha t}\cos(\omega_0 t)u(t) \leftrightarrow \frac{s+\alpha}{(s+\alpha)^2+\omega_0^2}$$

- In the same way:

$$\sin(\omega_0 t)u(t) \leftrightarrow \frac{\omega_0}{s^2+\omega_0^2}$$

$$\Rightarrow \quad e^{-\alpha t}\sin(\omega_0 t)u(t) \leftrightarrow \frac{\omega_0}{(s+\alpha)^2+\omega_0^2}$$

- Note that in the above examples, $s$ is simply replaced by $s+\alpha$.

# Complex-Frequency Differentiation

## Complex-Frequency Differentiation:

$$-tx(t) \leftrightarrow \frac{\mathrm{d}X(s)}{\mathrm{d}s}$$

Proof:

- $X(s)$ is given by:

$$X(s) = \int_{0^-}^{+\infty} x(t)e^{-st}\mathrm{d}t$$

- Consequently:

$$\begin{aligned}\frac{\mathrm{d}X(s)}{\mathrm{d}s} &= \frac{\mathrm{d}}{\mathrm{d}s}\int_{0^-}^{+\infty} x(t)e^{-st}\mathrm{d}t \\ &= \int_{0^-}^{+\infty} \frac{\mathrm{d}}{\mathrm{d}s}\left[x(t)e^{-st}\right]\mathrm{d}t \\ &= \underbrace{\int_{0^-}^{+\infty} -tx(t)e^{-st}\mathrm{d}t}_{=\mathcal{L}[-tx(t)]}\end{aligned}$$

# Integration

**Integration:**

$$\int_{0^-}^{t} x(\tau)\mathrm{d}\tau \leftrightarrow \frac{1}{s}X(s)$$

Proof:

- The integration property is easy to prove using the convolution property. $x(t) * u(t)$ is given by:

$$\begin{aligned} x(t) * u(t) &= \int_{-\infty}^{+\infty} x(\tau)u(t-\tau)\mathrm{d}\tau \\ &= \int_{0^-}^{t} x(\tau)\mathrm{d}\tau \end{aligned}$$

- The multiplication-convolution duality implies that:

$$x(t) * u(t) \;\leftrightarrow\; X(s)U(s) = \frac{1}{s}X(s)$$

# Time Differentiation (1)

**Time Differentiation:**

$$x'(t) = \frac{\mathrm{d}x(t)}{\mathrm{d}t} \leftrightarrow sX(s) - x(0^-)$$

Proof:

- $X(s)$ is given by:

$$X(s) = \int_{0^-}^{+\infty} x(t)e^{-st}\mathrm{d}t$$

- Evaluating the integral by parts $\int u\mathrm{d}v = uv - \int v\mathrm{d}u$ with $u = x(t)$ and $\mathrm{d}v = e^{-st}\mathrm{d}t$ results in:

$$\begin{aligned} X(s) &= \left[x(t)\left(-\frac{1}{s}e^{-st}\right)\right]_{0^-}^{+\infty} - \int_{0^-}^{+\infty}\left(-\frac{1}{s}e^{-st}\right)x'(t)\mathrm{d}t \\ &= \frac{1}{s}\left[x(0^-) - \lim_{t\to+\infty} x(t)e^{-st}\right] + \frac{1}{s}\underbrace{\int_{0^-}^{+\infty} x'(t)e^{-st}\mathrm{d}t}_{=\mathcal{L}[x'(t)]} \end{aligned}$$

# Time Differentiation (2)

- If $x(t)e^{-st} \to 0$ as $t \to +\infty$ (in other words, $x(t)$ grows no faster than an exponential function), the last equation implies that:

$$X(s) = \frac{1}{s}x(0^-) + \frac{1}{s}\mathcal{L}[x'(t)]$$

- Consequently:

$$\mathcal{L}[x'(t)] = sX(s) - x(0^-)$$

# Time Differentiation (3)

In the same way:

$$x''(t) = \frac{\mathrm{d}^2 x(t)}{\mathrm{d}t^2} \leftrightarrow s^2 X(s) - sx(0^-) - x'(0^-)$$

- In fact, since $x''(t) = \frac{\mathrm{d}x'(t)}{\mathrm{d}t}$:

$$\begin{aligned}\mathcal{L}\left[x''(t)\right] &= s\mathcal{L}\left[x'(t)\right] - x'(0^-) \\ &= s\left[sX(s) - x(0^-)\right] - x'(0^-) \\ &= s^2 X(s) - sx(0^-) - x'(0^-)\end{aligned}$$

By recursion, it can be proven that:

$$x^{(n)}(t) = \frac{\mathrm{d}^n x(t)}{\mathrm{d}t^n} \leftrightarrow s^n X(s) - \sum_{i=0}^{n-1} s^{n-1-i} x^{(i)}(0^-)$$

# Initial and Final Value Theorems

- Initial Value Theorem:

$$x(0^+) = \lim_{s \to +\infty} sX(s)$$

- Final Value Theorem:

$$\lim_{t \to +\infty} x(t) = \lim_{s \to 0} sX(s)$$

this property can be applied only if $\lim_{t \to +\infty} x(t)$ exists.

- For example: $x(t) = \cos(\omega_0 t)u(t) \leftrightarrow X(s) = \frac{s}{s^2+\omega_0^2}$:
  - $\lim_{s \to 0} sX(s) = 0$.
  - $\lim_{t \to +\infty} \cos(\omega_0 t)u(t) = \lim_{t \to +\infty} \cos(\omega_0 t)$ does not exist.
  - Consequently, we can not claim that $\lim_{t \to +\infty} \cos(\omega_0 t) = 0$ since the final value theorem can not be applied in this case.

# Example-7

Calculate the Laplace transform of $\text{rect}\left(t-\frac{1}{2}\right)$ (which is a causal signal):

$$x(t) = \text{rect}\left(t-\frac{1}{2}\right) = u(t) - u(t-1)$$

$$\Rightarrow X(s) = \frac{1}{s} - \frac{1}{s}e^{-s} = \frac{1-e^{-s}}{s}$$

- Deduce the Laplace transform of $\text{rect}\left(2t-\frac{1}{2}\right)$.
  - Applying the time scaling property:

$$\text{rect}\left(t-\frac{1}{2}\right) \leftrightarrow \frac{1-e^{-s}}{s}$$

$$t \to 2t \Rightarrow \text{rect}\left(2t-\frac{1}{2}\right) \leftrightarrow \frac{1}{2}\frac{1-e^{-\frac{s}{2}}}{\frac{s}{2}} = \frac{1-e^{-\frac{s}{2}}}{s}$$

# Example-8

Find the final value of the response $y(t)$ of a system whose transfer function is:

$$H(s) = \frac{s+3}{s^2+4s+5}$$

when this system is excited by $u(t)$ and $\delta(t)$ respectively.

- When the input is given by $x(t) = u(t)$:

$$Y(s) = X(s)H(s) = \frac{1}{s}H(s) = \frac{1}{s}\frac{s+3}{s^2+4s+5}$$

$$\Rightarrow \lim_{t\to+\infty} y(t) = \lim_{s\to 0} sY(s) = \lim_{s\to 0}\frac{s+3}{s^2+4s+5} = \frac{3}{5}$$

- When the input is given by $x(t) = \delta(t)$:

$$Y(s) = X(s)H(s) = H(s) = \frac{s+3}{s^2+4s+5}$$

$$\Rightarrow \lim_{t\to+\infty} y(t) = \lim_{s\to 0} sY(s) = \lim_{s\to 0}\frac{s^2+3s}{s^2+4s+5} = 0$$

# Example-9 (1)

Calculate the Laplace transform of $te^{-\alpha t}u(t)$:

- Applying the Complex-frequency differentiation:

$$-tx(t) \leftrightarrow \frac{\mathrm{d}X(s)}{\mathrm{d}s}$$

$$e^{-\alpha t}u(t) \leftrightarrow \frac{1}{s+\alpha}$$

$$\Rightarrow -te^{-\alpha t}u(t) \leftrightarrow \frac{\mathrm{d}}{\mathrm{d}s}\left[\frac{1}{s+\alpha}\right] = -\frac{1}{(s+\alpha)^2}$$

$$te^{-\alpha t}u(t) \leftrightarrow \frac{1}{(s+\alpha)^2}$$

# Example-9 (2)

- We can prove by induction that:

$$t^n e^{-\alpha t} u(t) \leftrightarrow \frac{n!}{(s+\alpha)^{n+1}}$$

where $n! = 1 \times 2 \times \cdots \times n$.

- For $\alpha = 0$:

$$t^n u(t) \leftrightarrow \frac{n!}{s^{n+1}}$$

  - For $n = 1$:

$$tu(t) \leftrightarrow \frac{1}{s^2}$$

- Replacing $n$ by $n-1$ results in:

$$\frac{t^{n-1}}{(n-1)!} e^{-\alpha t} u(t) \leftrightarrow \frac{1}{(s+\alpha)^n}$$

$$\frac{t^{n-1}}{(n-1)!} u(t) \leftrightarrow \frac{1}{s^n}$$